



## Alex Murray

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### EDUCATION

Bachelor of Science: Computer Science, University at Buffalo, Buffalo, New York, June 2026

### SKILLS & TOOLS

**Languages:** Java, Python, C++, Go, MIPS Assembly, Swift, HTML, CSS, TypeScript, OCaml  
**Data Management & Analytics:** MySQL, PostgreSQL, MongoDB, Firestore, Numpy, Matplotlib  
**Tools:** Jupyter, Visual Studio, JetBrains, Google Cloud, AWS, Clerk, Supabase, huggingface, Kaggle  
**Libraries and Frameworks:** Tailwind, FastAPI, OpenCV, React, shadcn, scikit-learn, Mediapipe

### COMPUTER SCIENCE PROJECTS

*Computer Vision Mouse, OpenCV, Mediapipe*

- Created a hands-free computer mouse interface using OpenCv and Mediapipe such that users can control their mouse pointer solely with their hands and a webcam
- Used dampening function to decrease latency, giving the user a smooth “mouse-like” experience

*Blackhole/Space Simulation, C++, OpenGL*

- Enhanced an existing project, updated physics/animations to help better understand the mysteries of our universe using 3D modeling

*AI Cartoon Generation, huggingface, AWS*

- Employed diffusion models on huggingface via AWS EC2 cloud compute to generate short snippets of cartoon images that are then compiled together to create short cartoons using python

*Automated Soccer Offside Detection, OpenCV, Numpy*

- Automated the process of determining offsides in soccer via video input and utilizing OpenCV, and Numpy to perform linear transformations that easily identify if a player is past the last defender

*Football Match Predictor, Numpy, scikit-learn*

- Fine-tuned scikit-learn and its machine learning algorithms to predict the outcomes English Premier League using Kaggle datasets and python

### WORK EXPERIENCE

*Data Science Intern, National Science Foundation, Buffalo, NY, June 2025 – August 2025*

- Re-developed the Popworld application to be used by thousands in the UB’s Evolutionary Biology course to simulate how evolutionary forces modify a population over the course of “n” generations
- Leveraged Google Cloud Run and Firestore to host the application in the cloud and simultaneously write and export the necessary user data from the Firestore database in a way that is accessible
- Utilized TailwindCSS, shadcn, and React functions to make a responsive and accessible frontend.

*Co-Founder, BetGenie, Buffalo, NY, May 2025 – Current*

- Collaborated on developing a sports-book discrepancy finder that gives users favorable bets using machine learning models through hugging face and scikit-learn
- Implemented user authentication and information system using Clerk and PostgreSQL via Supabase to keep track of user activity and payment information.
- Utilized AWS cloud compute to effectively train and run our prediction models, achieving an accuracy of 70%

### AWARDS & INVOLVEMENT

Welcome Weekend Leader, UB Orientation Office, August 2024

Intramural Soccer Referee, UB Recreation Intramural Sports Program, March 2023 – May 2025

University at Buffalo Dean’s List, Spring 2023- Fall 2024